

# Components of the Network



# Objectives:

- Identify the different components of the network.
- Define what intermediary devices are and its role on the network.
- Differentiate hub, switch and router.
- Define what network media is and its role on the the network.
- Differentiate UTP and STP cables.
- Understand how fiber optic cables work.

# Terms- Devices



Devices and media are the physical elements or hardware of the network.



Hardware is often the visible components of the network platform such as a laptop, a PC, a switch, or the cabling used to connect the devices.



In the case of wireless media, messages are transmitted through the air using invisible radio frequency or infrared waves.

# Terms

## PATH

The path that a message takes from source to destination can be as simple as a single cable connecting one computer to another or as complex as a network that literally spans the globe.

# Terms



## SERVICES



Services and processes are the communication programs, called software, that run on the networked devices.



A network service provides information in response to a request.



Services include many of the common network applications people use every day, like e-mail hosting services and web hosting services.



The background is a detailed, high-tech circuit board. It features a complex network of glowing blue and orange traces. On the right side, there's a large, dark rectangular area filled with glowing blue binary code (0s and 1s). The overall aesthetic is futuristic and digital.

# End Devices and Their Role on the Network



# End device

An end device refers to a piece of equipment that is either the source or the destination of a message on a network.

Network users usually only see and touch an end device, which is most often a computer. Another generic term for an end device that sends or receives messages is a host.

A host can be one of several pieces of equipment performing a wide variety of functions.

# End devices





# Examples of hosts and end devices are as follows:

- Computers, including workstations, laptops, and servers connected to a network
- Network printers
- Voice over Internet Protocol (VoIP) phones
- Cameras on a network, including webcams and security cameras
- Handheld devices such as PDAs and handheld scanners
- Remote monitoring stations for weather observation

## Examples of hosts and end devices are as follows:

An end user is a person or group using an end device. Not all end devices are operated by people all of the time, though.

For example, file servers are end devices that are set up by people but perform their tasks on their own.

Servers are hosts that are set up to store and share information with other hosts called clients.

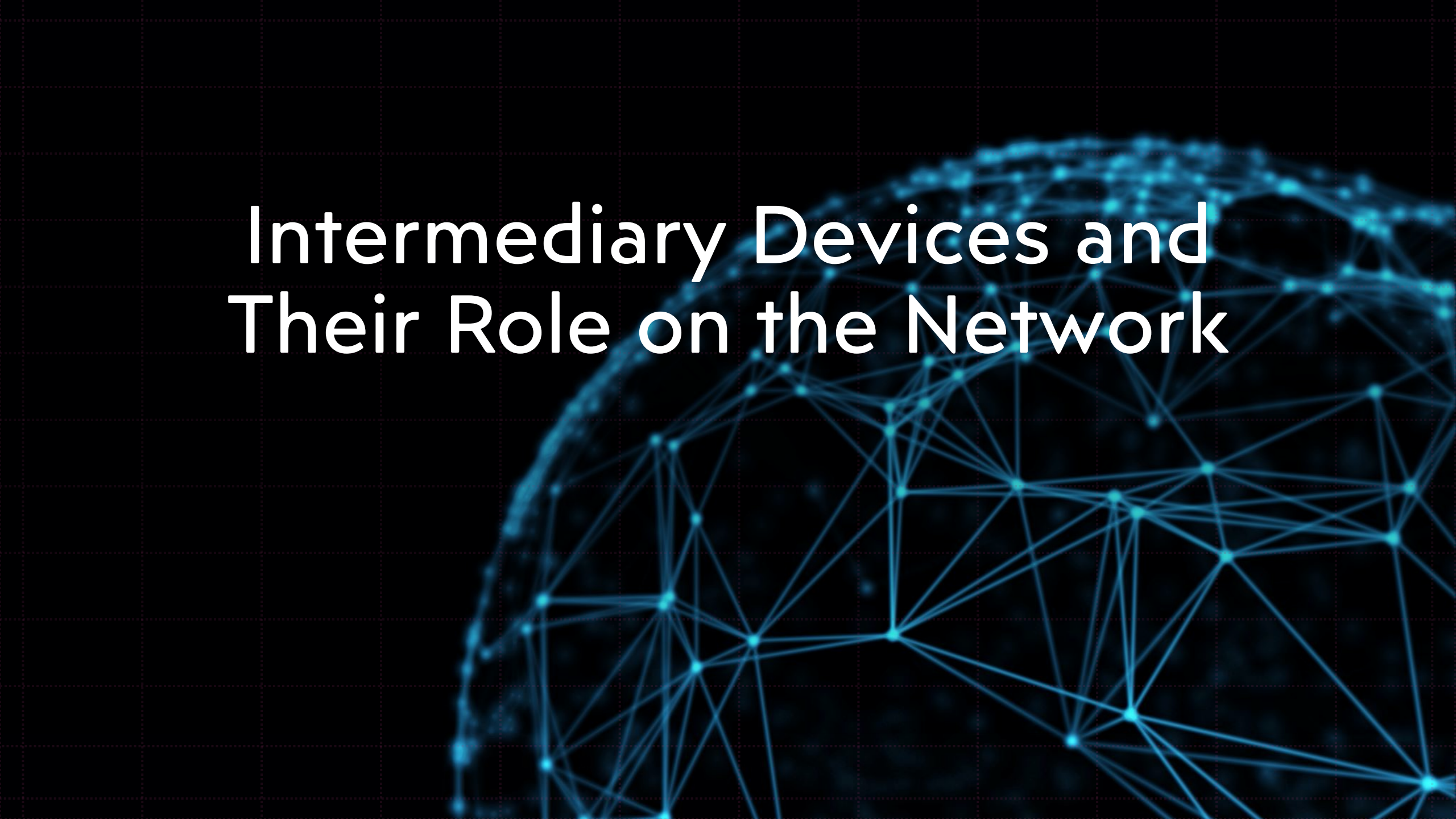
Clients request information and services, like e-mail and web pages, from servers, and servers reply with the requested information if they recognize the client.

# Examples of hosts and end devices are as follows:

- When hosts communicate with each other, they use addresses to find each other.
- The host address is a unique physical address used by hosts inside a local-area network (LAN), and when a host sends a message to another host, it uses the physical address of the destination device.



# Intermediary Devices and Their Role on the Network

The background of the slide features a complex, glowing blue network of interconnected nodes and lines, resembling a globe or a complex web. The nodes are small, bright blue dots, and the lines are thin, glowing blue lines that connect them in a dense, mesh-like pattern. The overall effect is a sense of a vast, interconnected digital space. The text is overlaid on this network, with the top half of the title positioned over the upper, more curved part of the network and the bottom half over the lower, more flat part.

# Intermediary Devices and their Role on the Network

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End devices are the hosts that initiate communications and are the ones that people are most familiar with.

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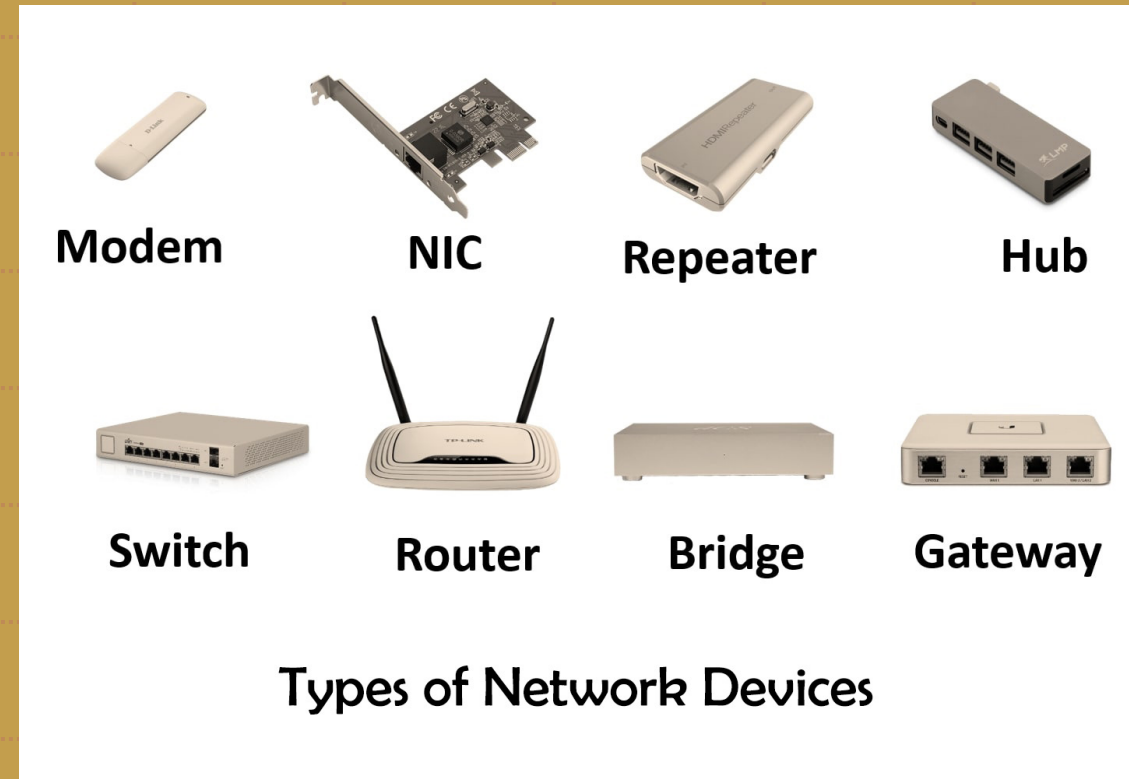
Intermediary devices connect the individual hosts to the network and can connect multiple individual networks to form an internetwork.

# Intermediary Devices

**Table 2-1** Intermediary Devices

| Device Type            | Description                                                                                           |
|------------------------|-------------------------------------------------------------------------------------------------------|
| Network access devices | Connect end users to their network. Examples are hubs, switches, and wireless access points.          |
| Internetwork devices   | Connect one network to one or more other networks. Routers are the main example.                      |
| Communication servers  | Route services such as IPTV and wireless broadband.                                                   |
| Modems                 | Connect users to servers and networks through telephone or cable.                                     |
| Security devices       | Secure the network with devices such as firewalls that analyze traffic exiting and entering networks. |

# Network Devices





# Intermediary Devices

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The management of data as it flows through the network is also a role of the intermediary devices.

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These devices use the destination host address, in conjunction with information about the network interconnections, to determine the path that messages should take through the network.

# Difference between a hub, switch and router



**Processes  
running on  
the  
intermediary  
network  
devices  
perform  
these  
functions:**

- Regenerate and retransmit data signals
- Maintain information about what pathways exist through the network and internetwork
- Notify other devices of errors and communication failures
- Direct data along alternate pathways when there is a link failure

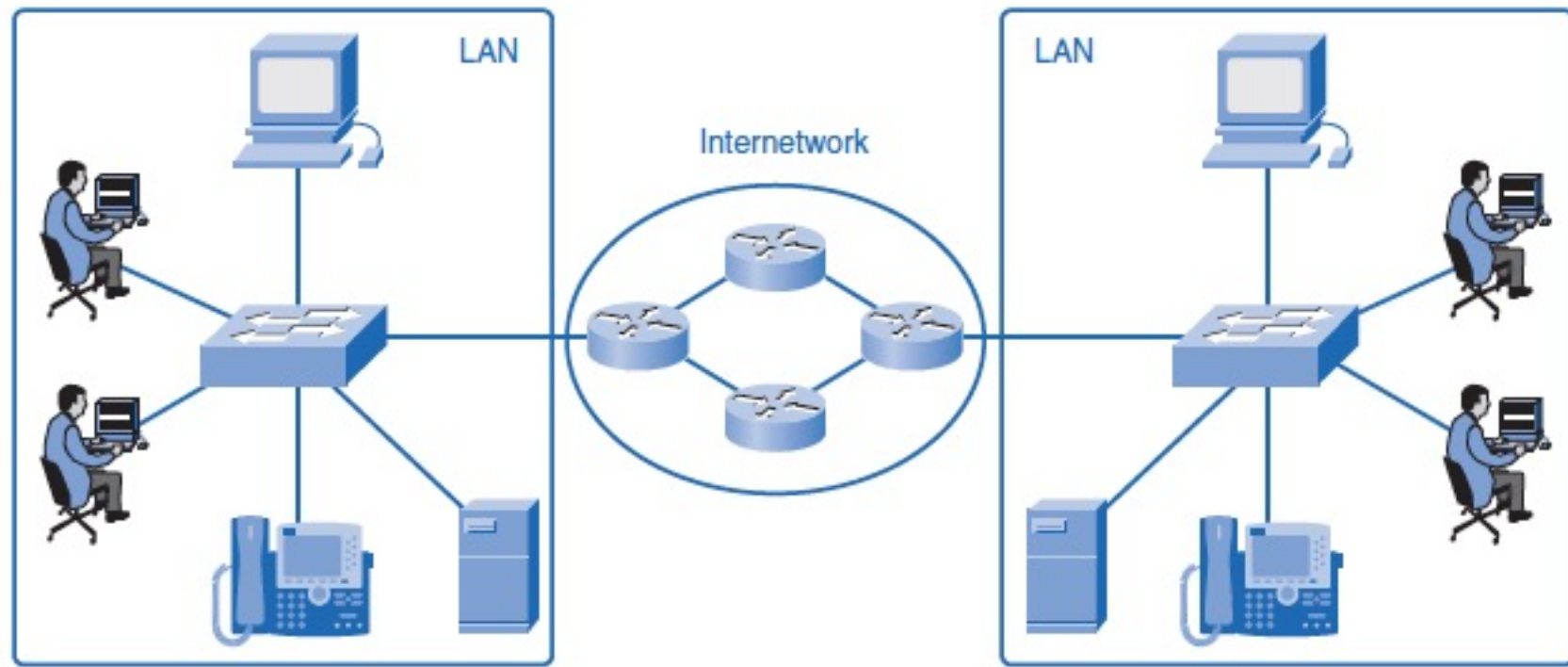
Processes running on the intermediary network devices perform these functions:

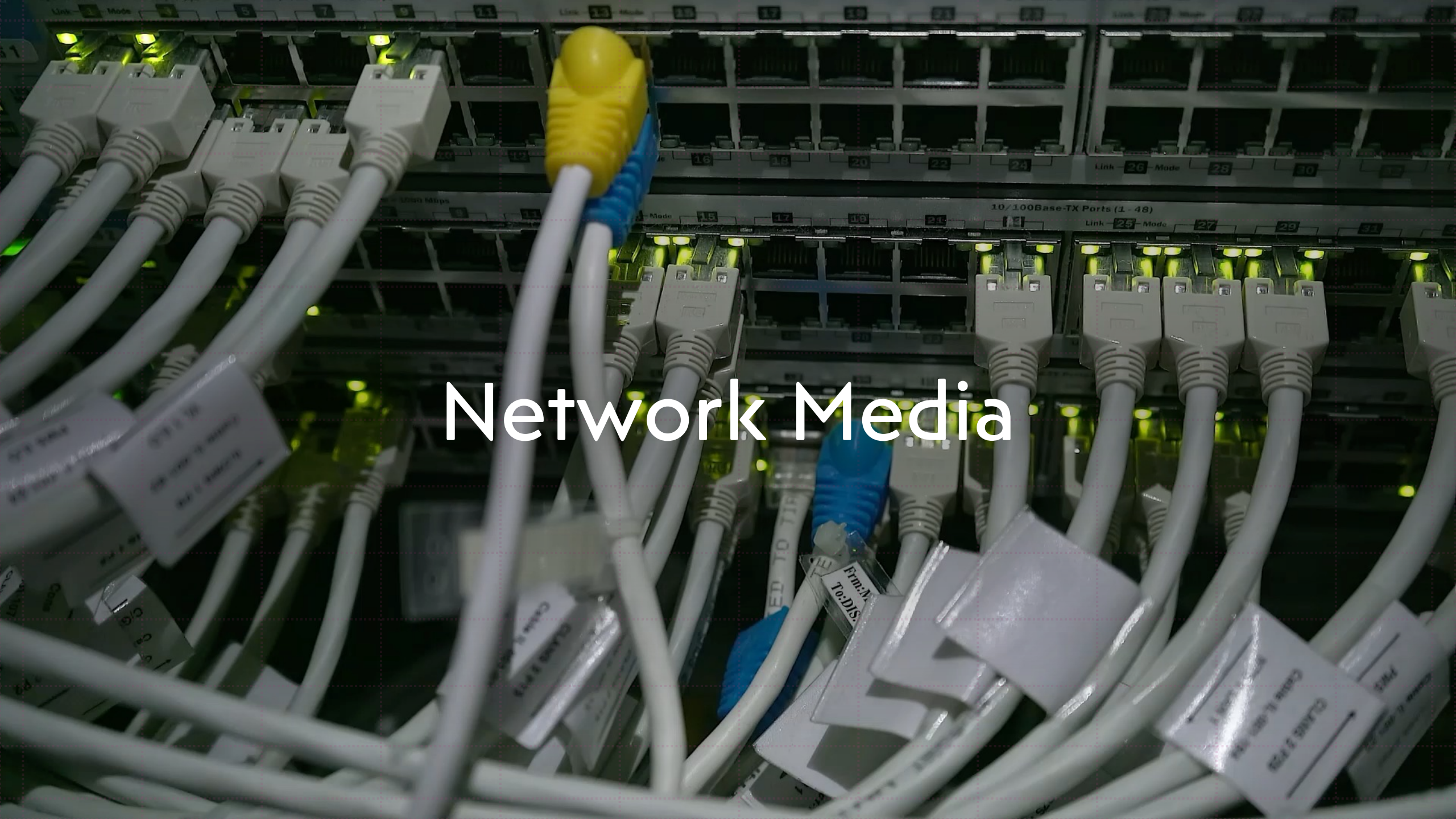
- Classify and direct messages according to quality of service (QoS) priorities
- Permit or deny the flow of data, based on security settings



# Intermediary Devices

**Figure 2-3** LANs Connected by Routers





# Network Media

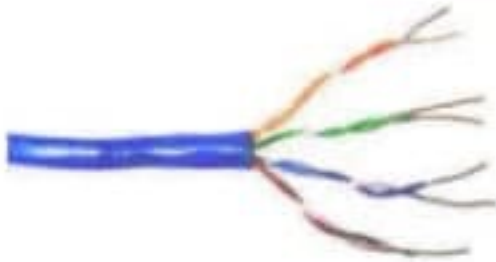


# Network Cables

Coaxial cable



Unshielded twisted pair



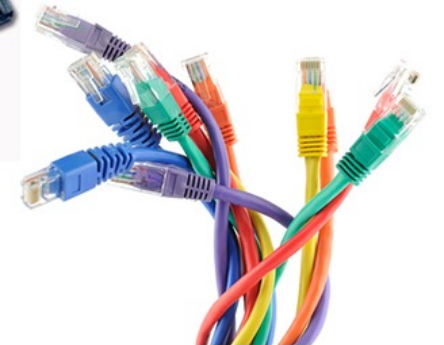
Fiber optic



UTP Cable



STP Cable



# How does data travel over the network?

- What are submarine cables?



# Network Media

- Communication across a network is carried on a medium.
- The medium provides the channel over which the message travels from source to destination.
- The three main types of media in use in a network are
  - Copper
  - Fiber-optic cable
  - Wireless



# Network Media

- Each of these media has vastly different physical properties and uses different methods to encode messages.
- Encoding messages refers to the way data is converted to patterns of electrical, light, or electromagnetic energy and carried on the medium.

# Network Media

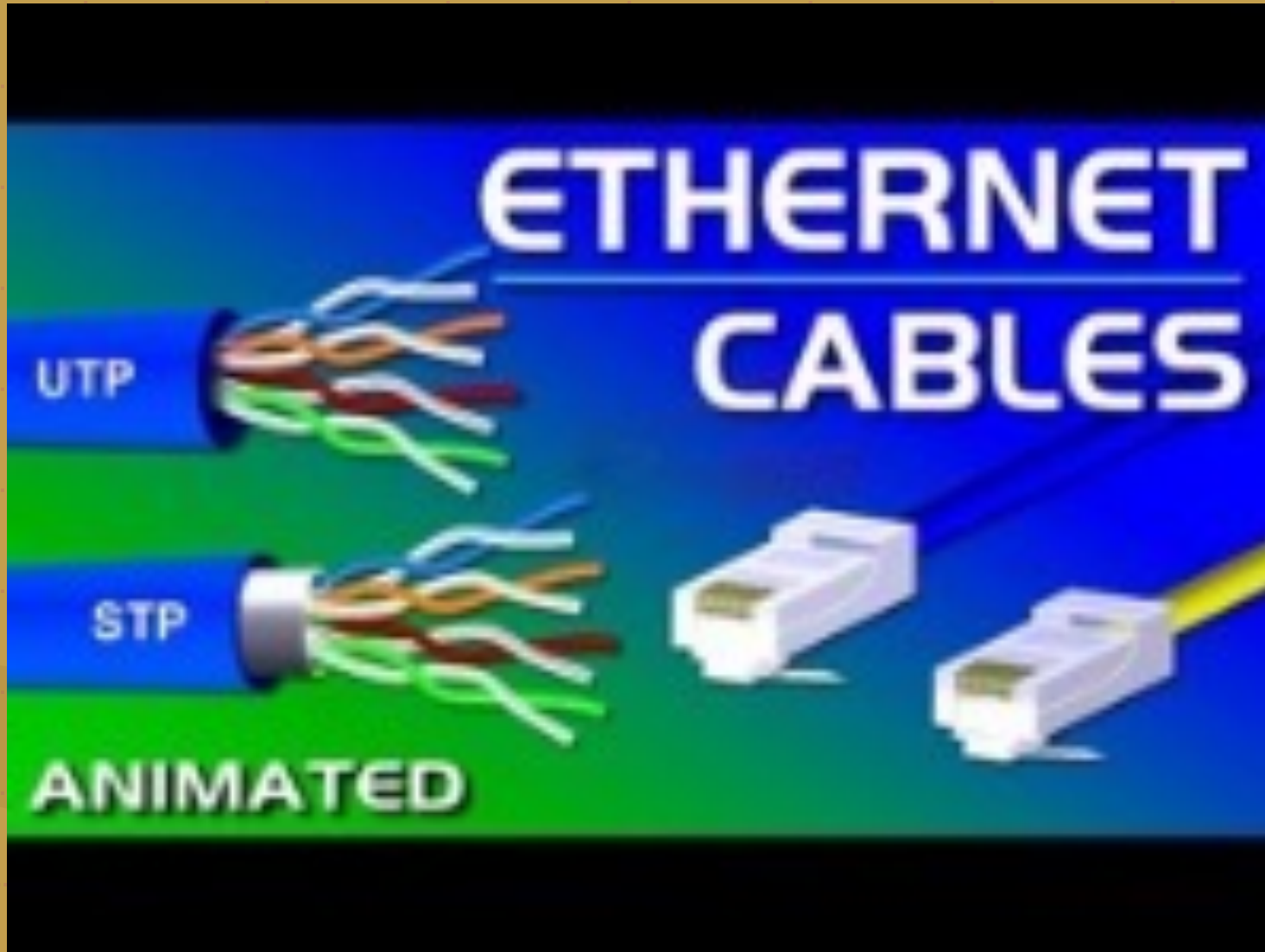
**Table 2-2**      Networking Media

| Media       | Example                                                                                       | Encoding              |
|-------------|-----------------------------------------------------------------------------------------------|-----------------------|
| Copper      | Twisted-pair cable usually used as LAN media                                                  | Electrical pulses     |
| Fiber-optic | Glass or plastic fibers in a vinyl coating usually used for long runs in a LAN and as a trunk | Light pulses          |
| Wireless    | Connects local users through the air                                                          | Electromagnetic waves |

# Network Media

- The differences in the media make each one ideal for different roles in networking situations.
- When choosing network media, administrators must consider the following:
  - The distance the media can carry the signal
  - The environment in which the media works
  - The bandwidth requirements for users
  - The cost of installation
  - The cost of connectors and compatible equipment

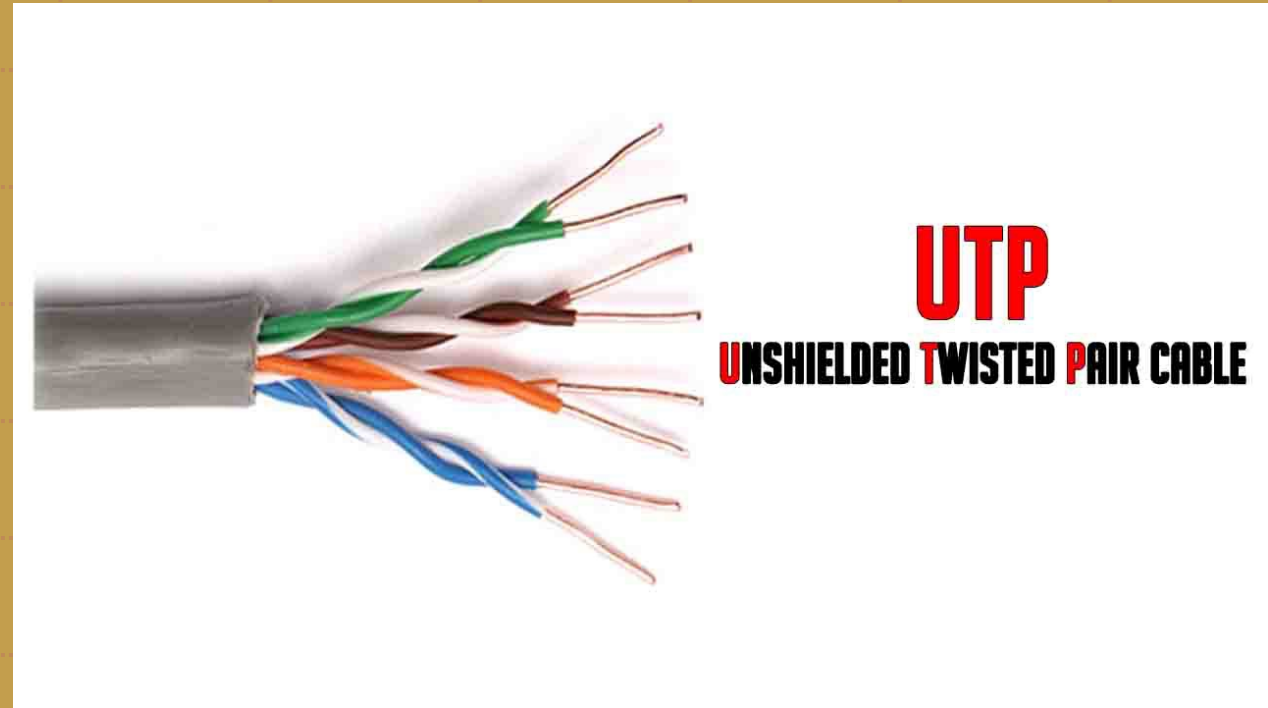
# Ethernet Cables





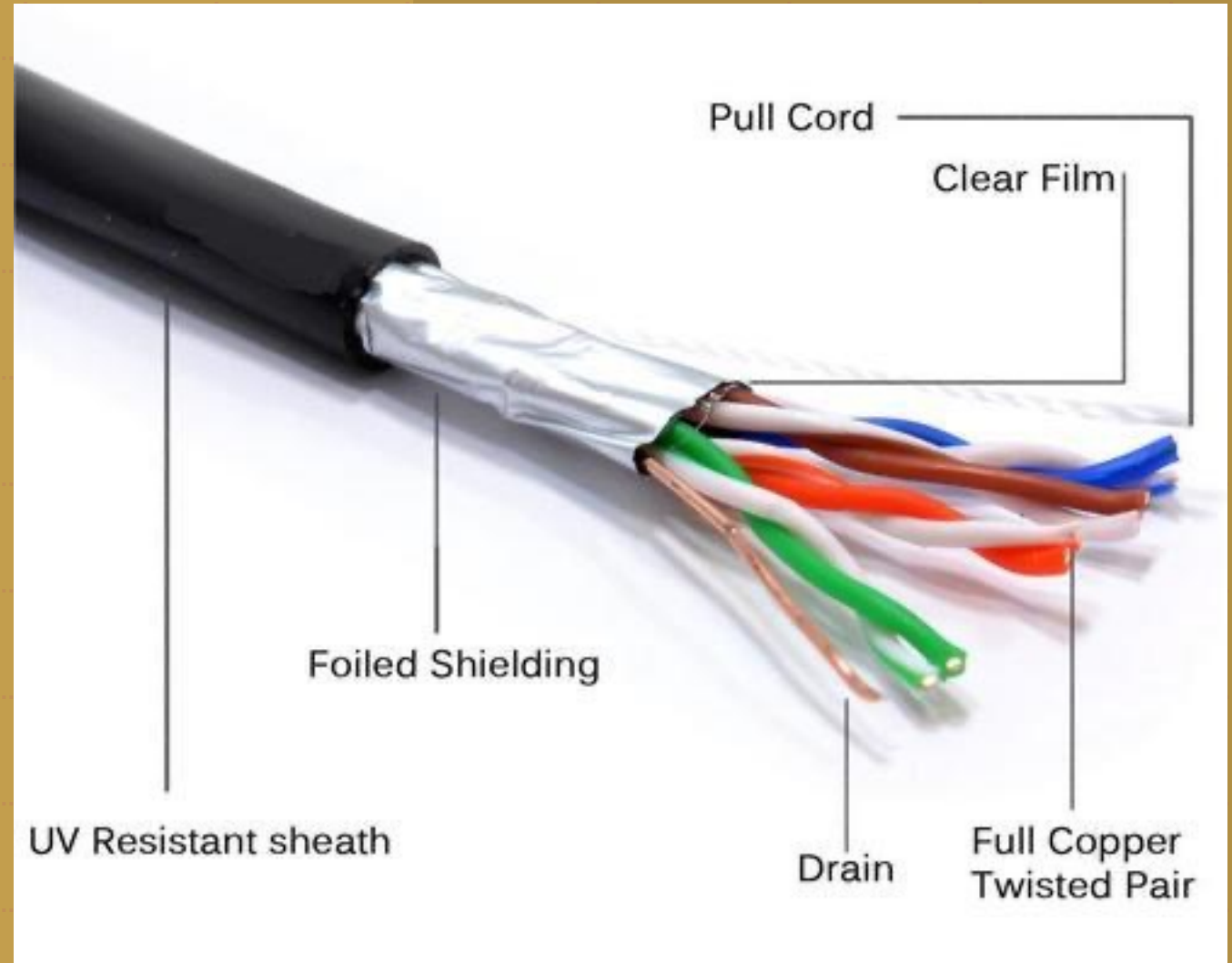
# Unshielded Twisted Pair (UTP)

- Unshielded Twisted Pair or UTP are twisted pair cables and are used to transmit both data and voice as their frequency range is suitable for transmission. UTPs are more cost effective and are not needed to be grounded.



# Shielded Twisted Pair (STP)

- Shielded Twisted Pair or STP are also a twisted pair cables but are required to be grounded, wants more maintenance, have high data transmission capacity and are more costly than UTP.



# Applications of STP

- Airports, medical centers, and factories often benefit from STP wiring. These facilities possess numerous machines that produce considerable amounts of interference.
- Shielded cables also come in handy when installers must run wires next to fluorescent lights, microwave ovens or powerful motors.
- These cables prove useful in outdoor settings as well. For instance, some people connect outdoor surveillance cameras to STP wiring. This protects the video feed from interference generated by vehicles and power lines. It also stops criminals from using jammers to interrupt communications between cameras and indoor monitors or recording equipment.

# UTP vs. STP

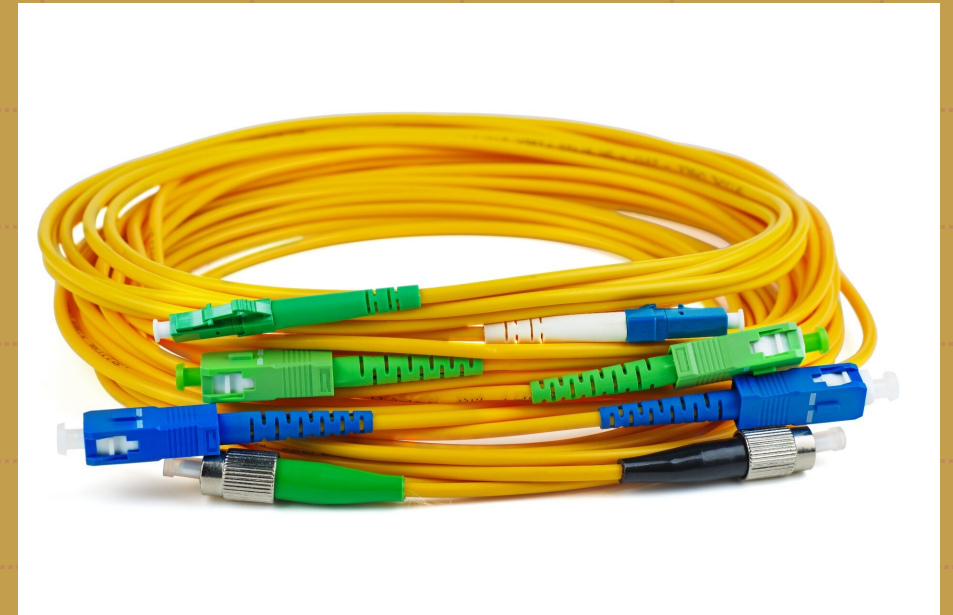
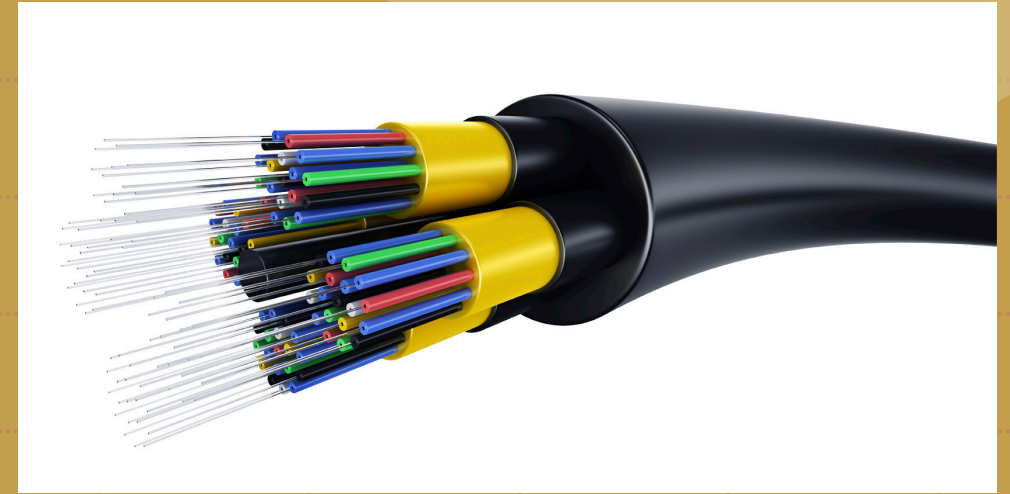
| Parameters           | UTP                                                          | STP                                                         |
|----------------------|--------------------------------------------------------------|-------------------------------------------------------------|
| Full Form            | Unshielded Twisted Pair                                      | Shielded Twisted Pair                                       |
| Structure            | cable with wires that are twisted together.                  | Twisted pair cable enclosed in foil / shield.               |
| Cost                 | Cheaper than STP                                             | Costlier than UTP                                           |
| Weight               | Lighter than STP                                             | Heavier than UTP                                            |
| Noise & interference | Prone to Noise and interference                              | Less prone to noise and interference                        |
| Data Speed           | Supports slower speed than on STP                            | Support higher speed than UTP                               |
| Grounding of cable   | Not required                                                 | Required                                                    |
| Target deployments   | Locations less prone to interference like offices and homes. | Locations prone to interference like factories and airports |

# What is a fiber optics cable?

- **Fiber optics is the technology used to transmit information as pulses of light through strands of fiber made of glass or plastic over long distances.**
- Optical fibers are about the diameter of a strand of human hair and when bundled into a fiber-optic cable, they're capable of transmitting more data over longer distances and faster than other mediums. It is this technology that provides homes and businesses with fiber-optic internet, phone and TV services.



# Fiber Optics Cable

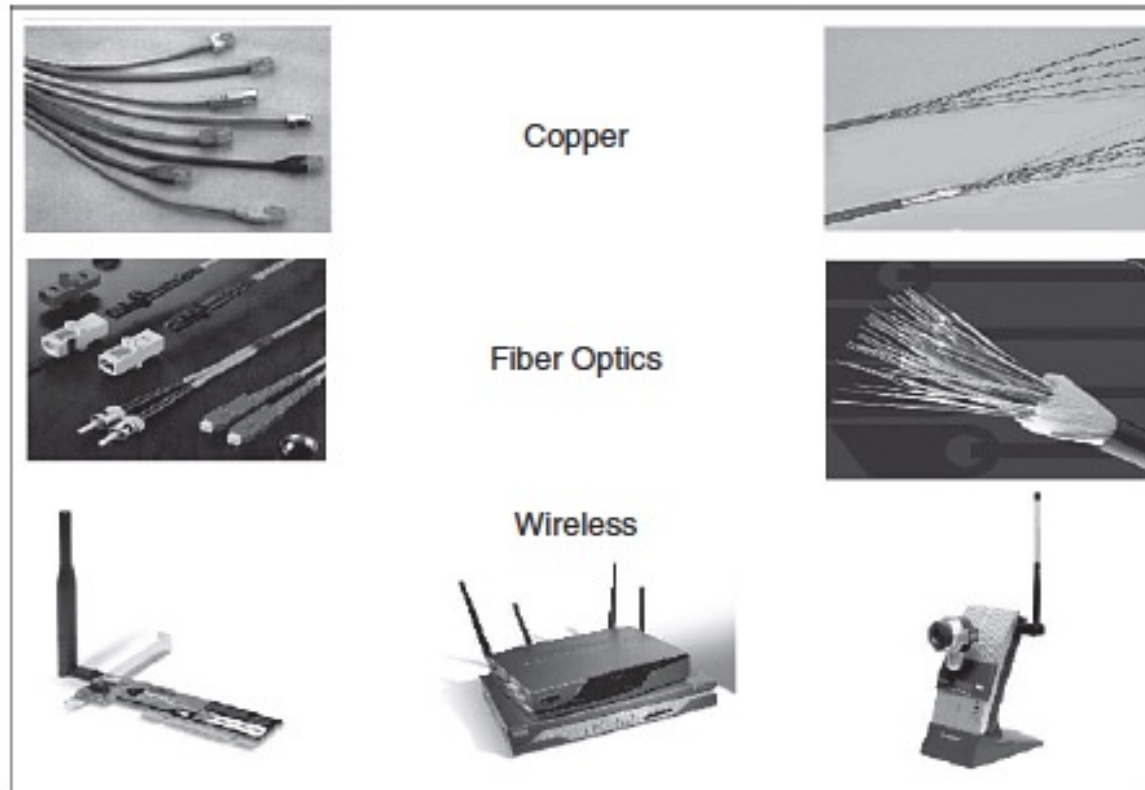


# Fiber Optics Cable



# Network Media

**Figure 2-4** Network Media



# In summary...

## End Devices



Desktop Computer



Laptop



Printer



IP Phone



Wireless Tablet



TelePresence  
Endpoint

## Intermediary Devices



Wireless Router



LAN Switch



Router



Multilayer Switch



Firewall Appliance

## Network Media



Wireless Media



LAN Media



WAN Media

# Helpful video links:

Difference between Hubs, Switches and Routers

[https://www.youtube.com/watch?v=Ofjsh\\_E4HfY&t=139s](https://www.youtube.com/watch?v=Ofjsh_E4HfY&t=139s)

Ethernet Cables

- [https://www.youtube.com/watch?v=\\_NX99ad2FUA&t=2s](https://www.youtube.com/watch?v=_NX99ad2FUA&t=2s)

Submarine Cables

- <https://www.youtube.com/watch?v=eTBLIYJSzdc>

Optical Fibers

- <https://www.youtube.com/watch?v=jZOg39v73c4>